REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

CLIFF
File: CLIFF.kicad_sch

MAIN BRUSH MOTOR
File: MAIN-BRUSH-MOTOR .kicad_sch

WHEEL MOTORs
File: WHEEL-MOTORs .kicad_sch

MAIN FAN
File: MAIN-FAN .kicad_sch

SIDE BRUSH MOTORs
File: SIDE-BRUSH-MOTORs .kicad_sch

BUMPER
Bumper Sensors
File: bumper_sensors.kicad_sch

MCU-STM32
File: STM32G473.kicad_sch

CARPET-SENSOR
File: Carpet-sensor.kicad_sch

WD
WATCHDOG
File: WATCHDOG.kicad_sch

LiDAR
File: LIDAR.kicad_sch

FRONT SENSORS
File: Front-Sensors.kicad_sch

USB-C POWER
USB Power
File: usb_power.kicad_sch

IMU
File: IMU-ICM-4267-P .kicad_sch

BUTTON-LEDs
File: BUTTON-LEDs .kicad_sch

POWER CONTROL
File: SYSTEM-POWER.kicad_sch

BATTERY CHARGER
File: Battery-Charger.kicad_sch

Sheet: /
File: Main.kicad_sch
Title:
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/17

5V USB, SENSORS, LIDAR POWER RAIL
– set 5V_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode; Fsw 418 kHz (RT 232k)

U22
TPS54560BDDAR
BOOT
VIN
EN
RT/CLK
PAD
SW
GND
COMP
FB
10 R146
C119 100nF/50V
L3
CYA0630-3_3UH
D43
SS56
VBAT-14V4
PWR_FLAG
VSENS_5V_USB_END
VSENS-5V-USB
D46
YLED0603B
C115 10uF/50V
C116 10uF/50V
C117 100nF/50V
R206 232k
R168 100k
PWR_FLAG
R147 4.7k
C120 47nF/50V
R148 12k
R149 2.2k
C121 10uF/10V
C122 10uF/10V
C147 10uF/10V
1k
R89

MOTOR POWER RAIL 14.4V

– on/off motor power, no soft ramp
– set VMOT-14V4-EN LOW to enable power
  – floating (MCU flash/reset) disables power
– WD_OK LOW shuts power off

VMOT-14V4: Q1 + Q37 AO4407C in parallel
– ~7.5 A normal (with the MSD-G fan) is ~0.25 W each
– short stall peaks (~13 A) fine with copper under the drain
– Vgs +/-25 V, no gate divider needed

VBAT-14V4
Q1
AO4407C
PWR_FLAG
VMOT-14V4
R165 10k
R36 10
VMOT-SW-G
C62
10uF/25V
D37
SS34
Q37
AO4407C
R222 10
VMCU-3V3
10k
R184
Q19
AO3400
WD_OKD
VMOT-14V4-END

3.3V MCU POWER RAIL

– POWER_EN is 14.4V
– set POWER_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode: fsw 418 kHz (RT 232k)
– EN abs max 8.4 V: R207/R208 on U21

U21
TPS54560BDDAR
BOOT
VIN
EN
RT/CLK
PAD
SW
GND
COMP
FB
10 R141
C111 100nF/50V
L2
CYA0630-3_3UH
D42
SS56
VBAT-14V4
R207 100k
R208 22.1k
C106 10uF/50V
C95 10uF/50V
C102 100nF/50V
R205 232k
C112 47nF/50V
R143 4.7k
R145 10k
R144 232k
C113 10uF/10V
C114 10uF/10V
C146 10uF/10V
PWR_FLAG
VMCU-3V3
D44
YLED0603B
1k
R14

14.5V ULTRASONIC CARPET SENSOR POWER

L4
XRNR4030-10UH_M
D4
SS34_C8678
PWR_FLAG
VUSONIC-14V4
VSENS-5V-USB
U5
NC
IN
EN
SW
GND
FB
MT3608
R160 232k
R159 10k
C149 10uF/10V
C148 10uF/10V
C150 10uF/25V
C151 10uF/25V

VUSONIC-14V4 = 0.6V x (1 + 232k/10k) ~= 14.5V

STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1– Hold EN to HIGH after buttone pressed
2– OFF board

VBAT-14V4
Q13
AO4407C
POWER-EN
R16 100k
R67 10k
C14 1uF/25V
R171 100k
PWR-SW-G
POWER-BTN
C22 1uF/25V
VMCU-3V3
R173 10k
PWR-BTN-SENSE
D26
1N4148
VMCU-3V3
R172 10k
R190 1k
Q23
AO3400
STM-PWR-CTRL
PWR-LATCH
VMCU-3V3
C173 100nF/50V
VMCU-3V3
R193 1M
R191 100k
R192 47k
Q29
AO3400
C172 10uF/10V
U23
74LVC1G17
Q30
AO3400

Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR-LATCH low), Firmware turns off at 3 s.
While charging, the rail-hold (Q31 on BATTERY-CHARGER, from DC_IN) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR-SW-G low).

VMCU-3V3
Q16
AO3401
VSENS-3V3
PWR_FLAG
100k
R99
1uF/25V
C176
D45
YLED0603B
1k
R98
VSENS_3V3_END
10k
R195
C64
100nF/50V

3.3V SENSORS POWER RAIL

– C176 keeps Q16 off when power switches ON
– C64 soft-ramps power
– set VSENS_3V3_EN LOW to enable sensors power

MCU VBAT
– Low quiescent current 2.5µA
– Zener diode shuts off when main battery is low

VBAT-14V4
D2
ZMM9V1_C2891750
U16
HT7533-1_C14289
Vout
Vin
GND
PWR_FLAG
VMCU-BAT-3V3
C140 10uF/25V
C141 10uF/25V

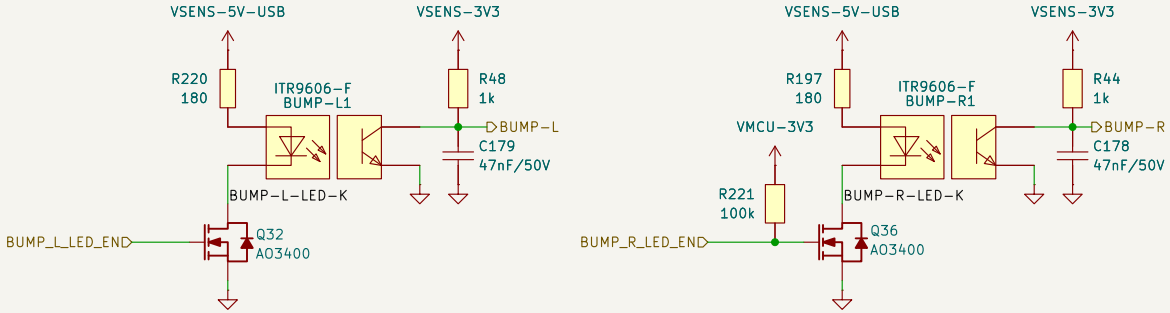
PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.

MP1
MP2
MP3
MP4
MP5
MP6

Sheet: /POWER/
File: SYSTEM-POWER.kicad_sch

Title:

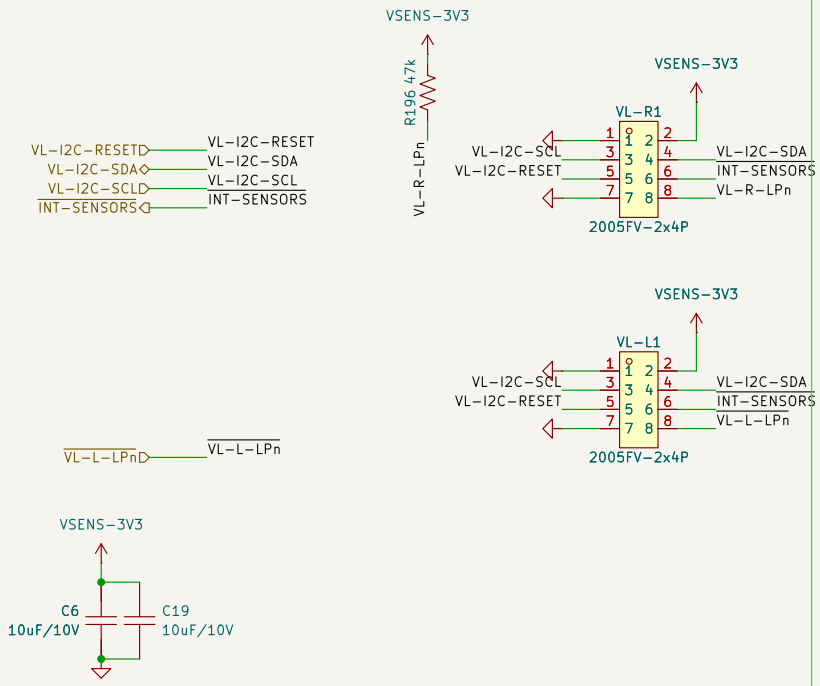
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 2/17



- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range
  1k keeps photo transistor output out of saturation
  so PA4/PA5 (ADC2) see the transition (ITR9606 IC(ON) 0.5-10 mA)
- R221 enables MCU USB-C Rd on both CC1 and CC2 during MCU *reset*,
  so the phone detects the robot whichever way the cable is plugged in

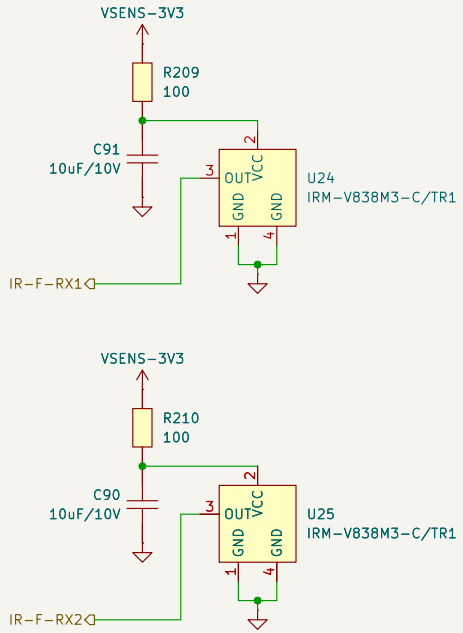
Sheet: /Bumper Sensors/ File: bumper_sensors.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/17

2x VL53L7 ToF BOARDS



2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. ~ 90 deg FoV

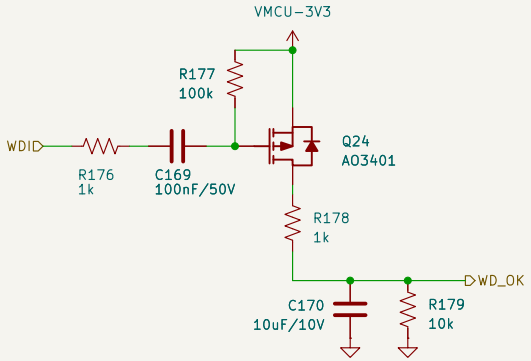


Sheet: /FRONT SENSORS/
File: Front-Sensors.kicad_sch

Title:

Size: USLetter | Date: | **Rev:**

KiCad E.D.A. 10.0.4 | Id: 4/17



RC CHARGE-PUMP WATCHDOG

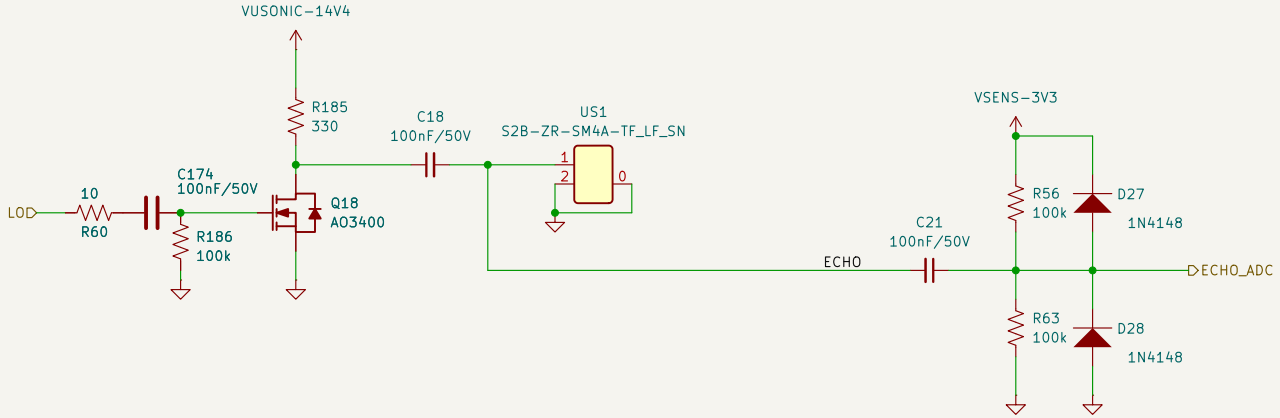
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/17

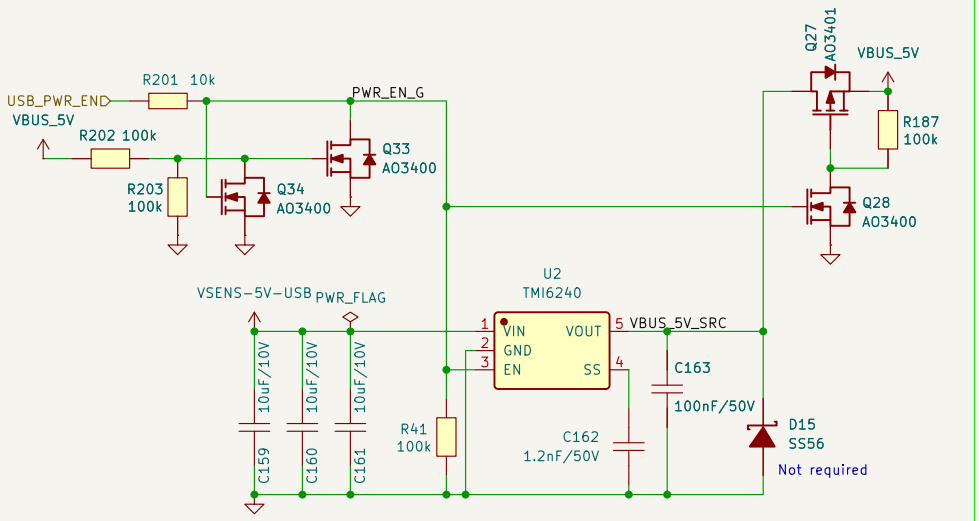
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



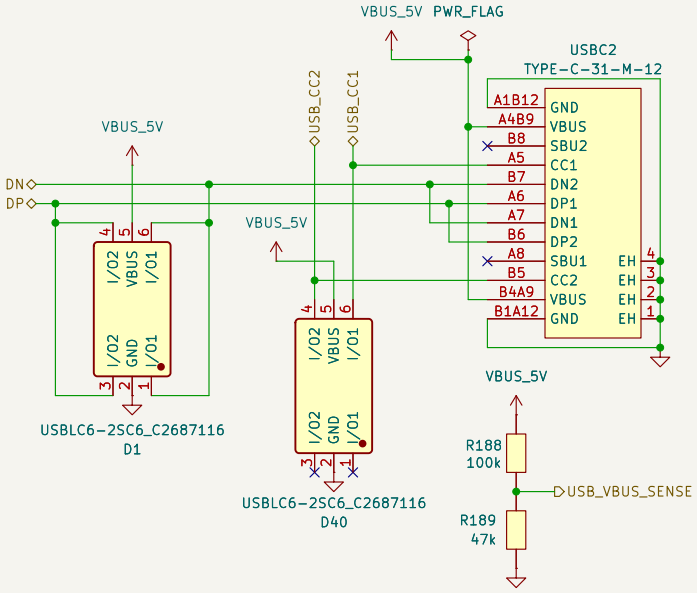
Sheet: /CARPET-SENSOR/	
File: Carpet-sensor.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 6/17

USB-C for SMARTPHONE



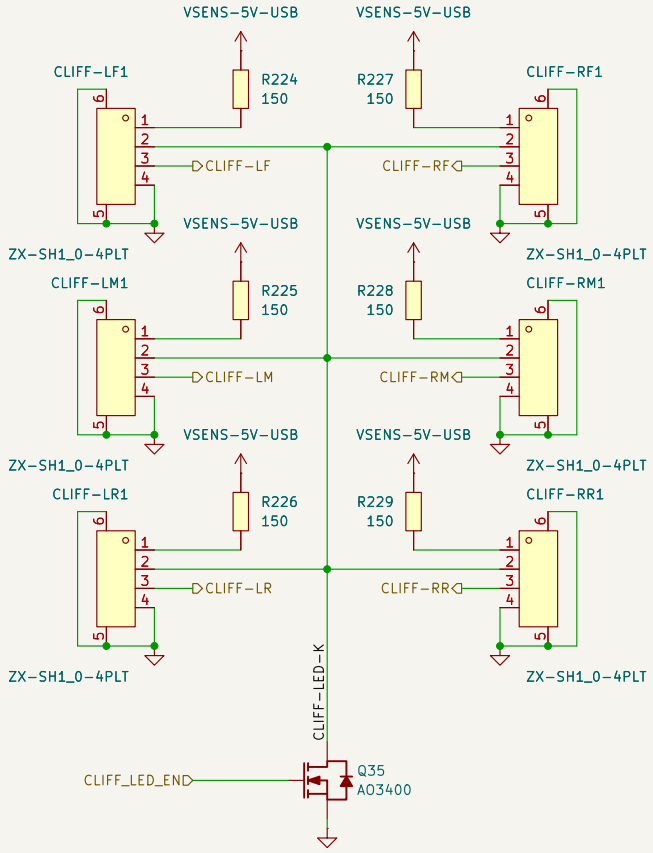
VBUS_5V lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS_5V is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS_5V; once on, Q34 latches it.

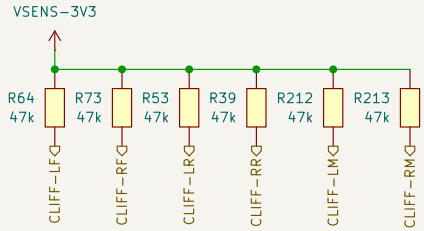
PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS_5V from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); USB_PWR_EN turns on TMI6240 and Q27 together.
TMI6240 now runs from VSENS-5V-USB (U25 buck removed).



USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default.

Sheet: /USB Power/	
File: usb_power.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 7/17



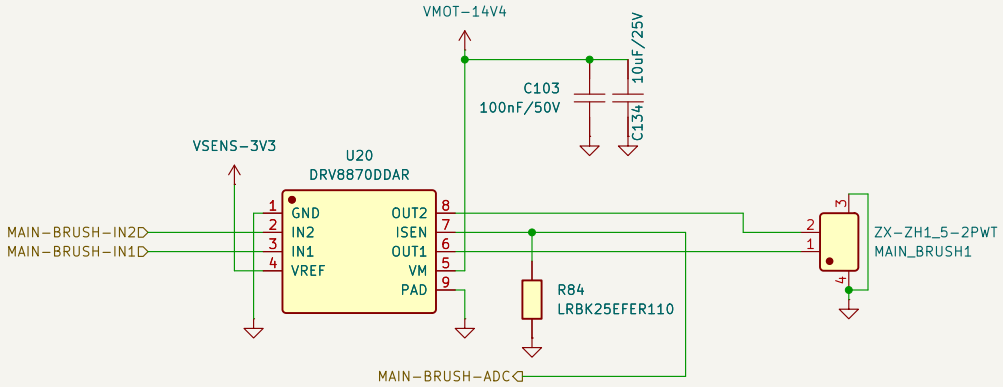


- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light
- possible TODO - cliff LEDs 2x or 3x in series, save resistors
  - worsens power rail noise rejection
  - ADC fw could reject noise by correlating the 6 readings
- 0.76W power consumption when enabled continuously
  - drive at ~10% duty cycle to save power
- tweak output pullup
  - a low resistor value can saturating the photo transistor
  - 47K pullup makes for 1ms output settling time
  - can use 10k for faster settling, to reduce duty cycle further
- combine low duty with fast duty LED cycling
  - fast (~10x photo transistor settling time) reduces LED average radiance to avoid photo transistor saturation
- check sensor faults - is output 3.3V or floating?
  - normal no-cliff condition - photo transisstor gets LED light Vary LED current and check photo-transistor voltage.

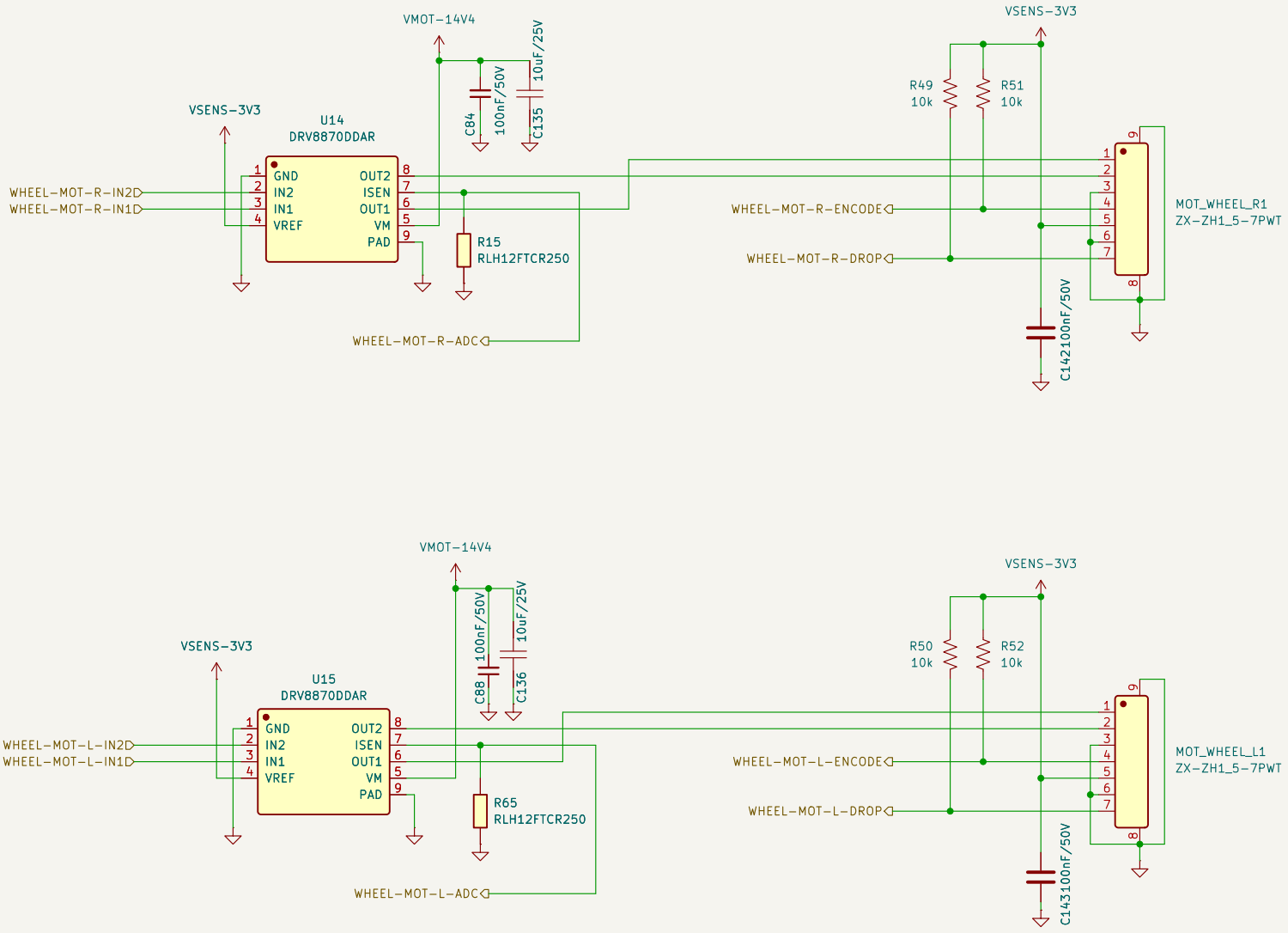
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File: CLIFF.kicad_sch

Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 19/17



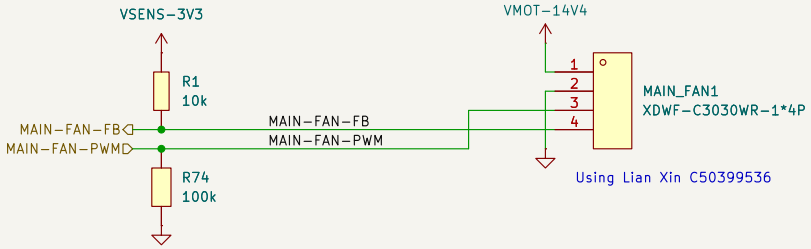
Sheet: /MAIN-BRUSH MOTOR/	
File: MAIN-BRUSH-MOTOR .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 22/17



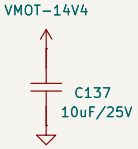
Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch

Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 23/17



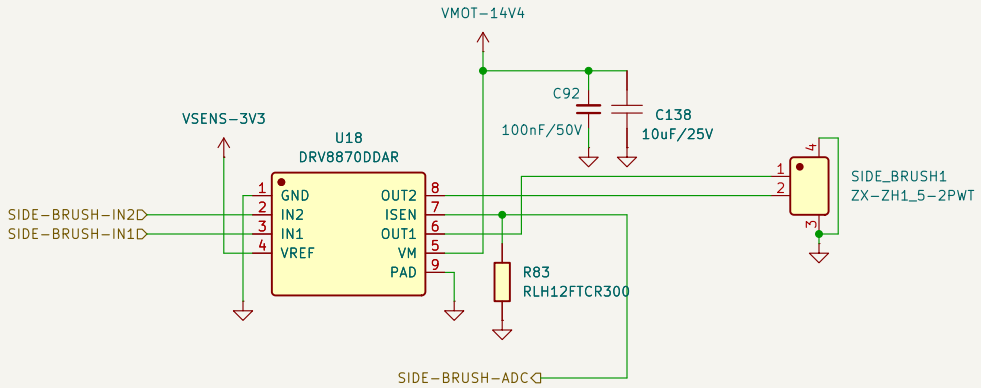
- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB



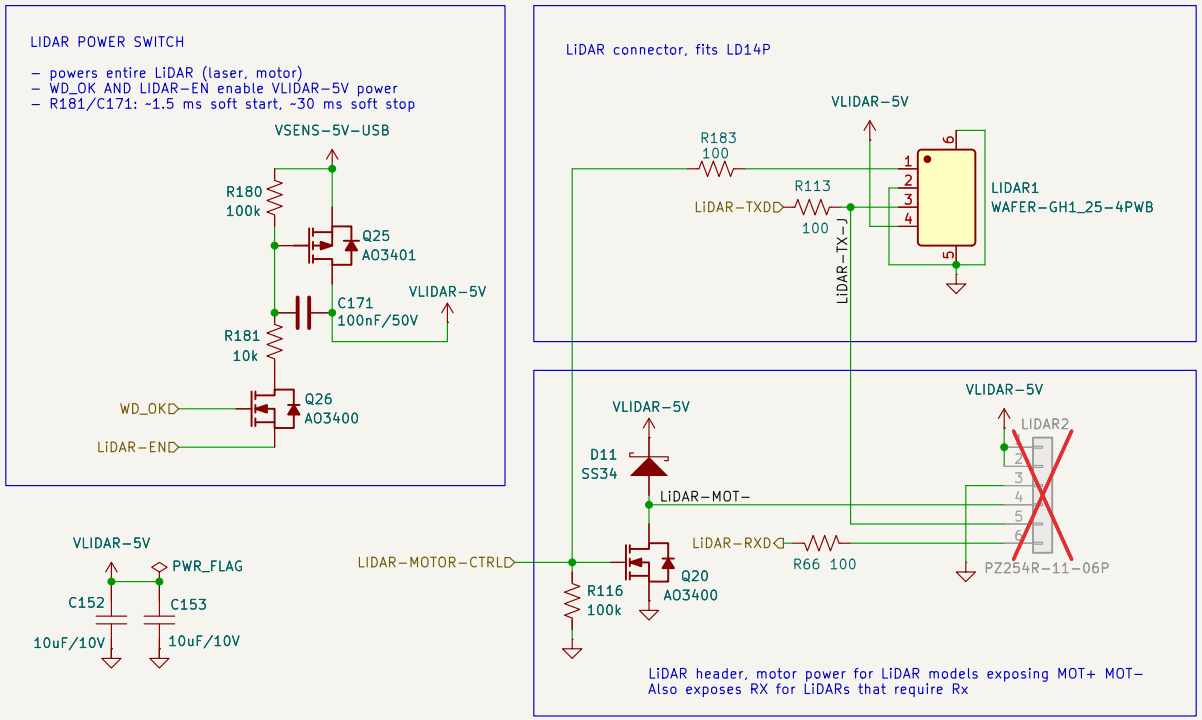
Sheet: /MAIN-FAN/
File: MAIN-FAN .kicad_sch

Title:

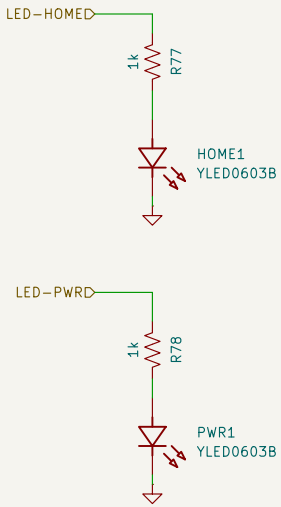
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 24/17

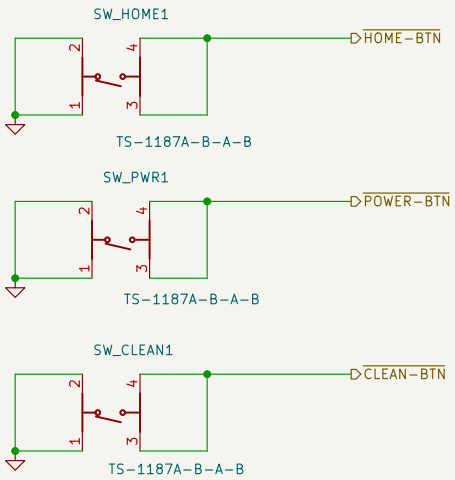


Sheet: /SIDE-BRUSH/		
File: SIDE-BRUSH-MOTORs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 25/17

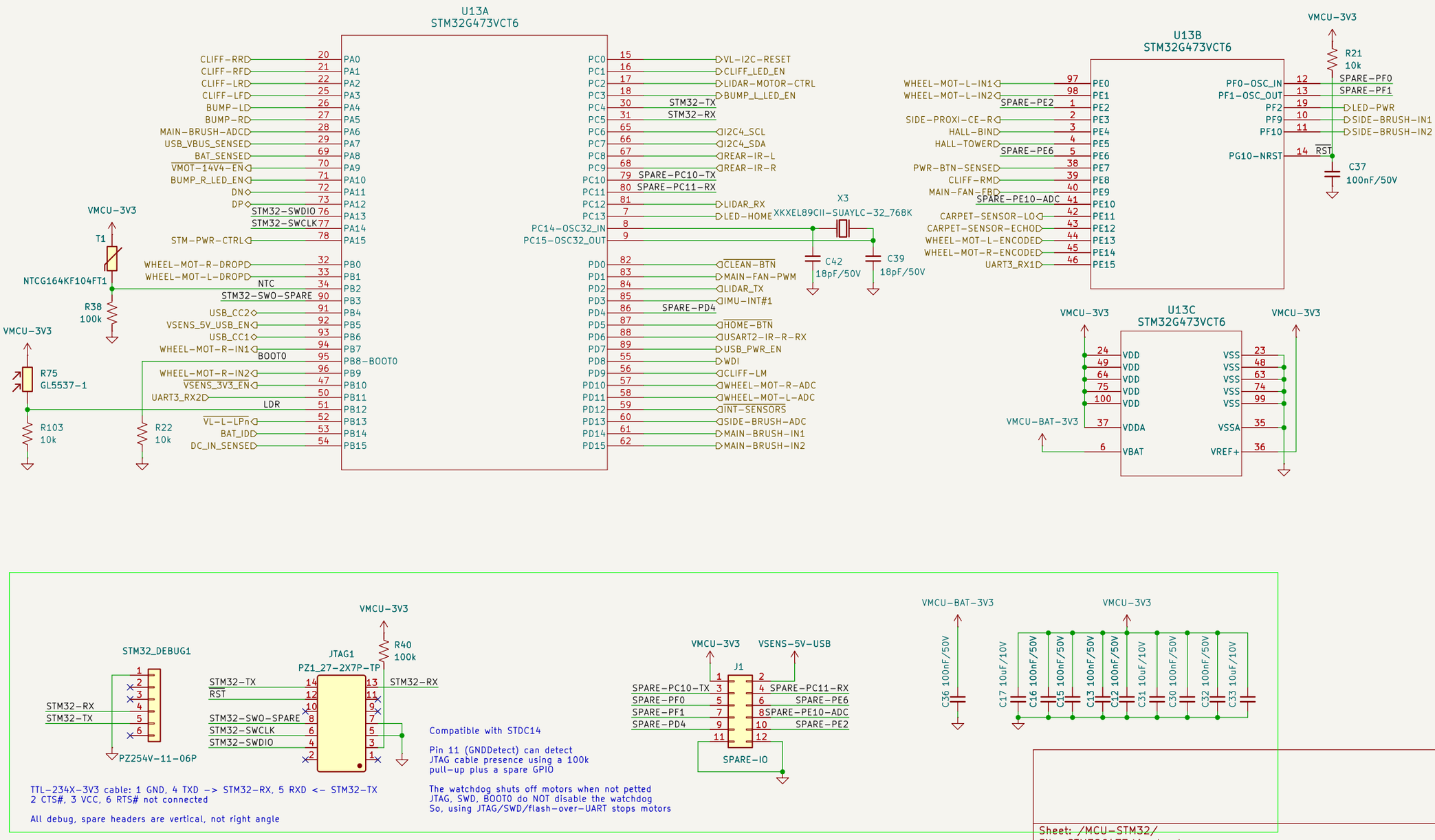


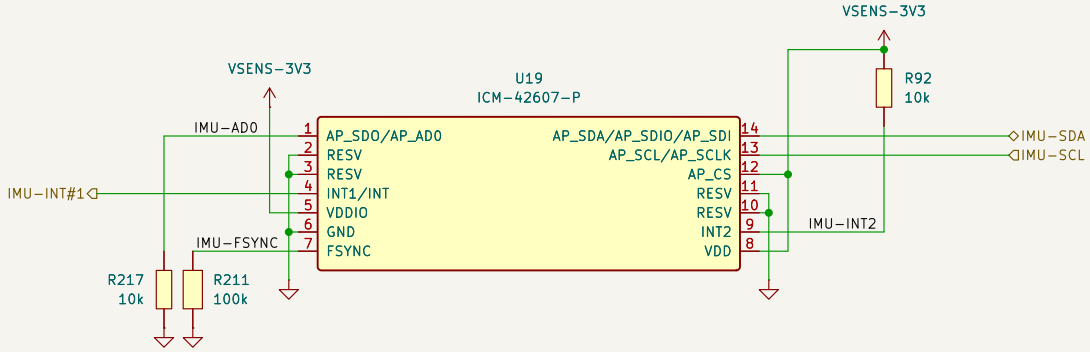
Sheet: /LIDAR/		
File: LIDAR.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 26/17



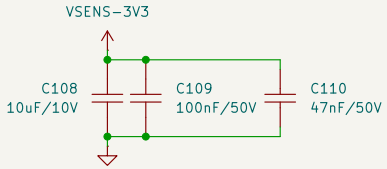


Sheet: /BUTTON-LED/		
File: BUTTON-LEDs .kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
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U13A
STM32G473VCT6
CLIFF-RR 20 PA0
CLIFF-RF 21 PA1
CLIFF-LR 22 PA2
CLIFF-LF 25 PA3
BUMP-LD 26 PA4
BUMP-RD 27 PA5
MAIN-BRUSH-ADC 28 PA6
USB_VBUS_SENSED 29 PA7
BAT_SENSED 69 PA8
VMOT-14V4-EN 70 PA9
BUMP_R_LED_EN 71 PA10
DN 72 PA11
DP 73 PA12
STM32-SWDIO 76 PA13
STM32-SWCLK 77 PA14
STM-PWR-CTRL 78 PA15
WHEEL-MOT-R-DROP 32 PB0
WHEEL-MOT-L-DROP 33 PB1
NTC 34 PB2
STM32-SWO-SPARE 90 PB3
USB_CC2 91 PB4
VSENS_5V_USB_EN 92 PB5
USB_CC1 93 PB6
WHEEL-MOT-R-IN1 94 PB7
BOOT0 95 PB8-BOOT0
WHEEL-MOT-R-IN2 96 PB9
VSENS_3V3_EN 47 PB10
UART3_RX2 50 PB11
LDR 51 PB12
VL-L-LPn 52 PB13
BAT_ID 53 PB14
DC_IN_SENSED 54 PB15
PC0 15 VL-I2C-RESET
PC1 16 CLIFF_LED_EN
PC2 17 LIDAR-MOTOR-CTRL
PC3 18 BUMP_L_LED_EN
PC4 30 STM32-TX
PC5 31 STM32-RX
PC6 65 I2C4_SCL
PC7 66 I2C4_SDA
PC8 67 REAR-IR-L
PC9 68 REAR-IR-R
PC10 79 SPARE-PC10-TX
PC11 80 SPARE-PC11-RX
PC12 81 LIDAR_RX
PC13 7 LED-HOME
PC14-OSC32_IN 8
PC15-OSC32_OUT 9
PD0 82 CLEAN-BTN
PD1 83 MAIN-FAN-PWM
PD2 84 LIDAR_TX
PD3 85 IMU-INT#1
PD4 86 SPARE-PD4
PD5 87 HOME-BTN
PD6 88 USART2-IR-R-RX
PD7 89 USB_PWR_EN
PD8 55 WDI
PD9 56 CLIFF-LM
PD10 57 WHEEL-MOT-R-ADC
PD11 58 WHEEL-MOT-L-ADC
PD12 59 INT-SENSORS
PD13 60 SIDE-BRUSH-ADC
PD14 61 MAIN-BRUSH-IN1
PD15 62 MAIN-BRUSH-IN2
X3
XKXEL89CII-SUAYLC-32_768K
C42 18pF/50V
C39 18pF/50V
VMCU-3V3
T1
NTCG164KF104FT1
R38 100k
R75 GL5537-1
R103 10k
R22 10k
U13B
STM32G473VCT6
WHEEL-MOT-L-IN1 97 PE0
WHEEL-MOT-L-IN2 98 PE1
SPARE-PE2 1 PE2
SIDE-PROXI-CE-R 2 PE3
HALL-BIN 3 PE4
HALL-TOWER 4 PE5
SPARE-PE6 5 PE6
PWR-BTN-SENSED 38 PE7
CLIFF-RM 39 PE8
MAIN-FAN-FB 40 PE9
SPARE-PE10-ADC 41 PE10
CARPET-SENSOR-LO 42 PE11
CARPET-SENSOR-ECHO 43 PE12
WHEEL-MOT-L-ENCODER 44 PE13
WHEEL-MOT-R-ENCODER 45 PE14
UART3_RX1 46 PE15
PF0-OSC_IN 12 SPARE-PF0
PF1-OSC_OUT 13 SPARE-PF1
PF2 19 LED-PWR
PF9 10 SIDE-BRUSH-IN1
PF10 11 SIDE-BRUSH-IN2
PG10-NRST 14 RST
R21 10k
C37 100nF/50V
U13C
STM32G473VCT6
24 VDD
49 VDD
64 VDD
75 VDD
100 VDD
37 VDDA
6 VBAT
VMCU-BAT-3V3
23 VSS
48 VSS
63 VSS
74 VSS
99 VSS
35 VSSA
36 VREF+
STM32_DEBUG1
PZ254V-11-06P
JTAG1
PZ1_27-2X7P-TP
R40 100k
STM32-TX
RST
STM32-SWO-SPARE
STM32-SWCLK
STM32-SWDIO
STM32-RX
Compatible with STDC14
Pin 11 (GNDDetect) can detect JTAG cable presence using a 100k pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
All debug, spare headers are vertical, not right angle
J1
VSENS-5V-USB
SPARE-PC10-TX
SPARE-PF0
SPARE-PF1
SPARE-PD4
SPARE-PC11-RX
SPARE-PE6
SPARE-PE10-ADC
SPARE-PE2
SPARE-IO
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch
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IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

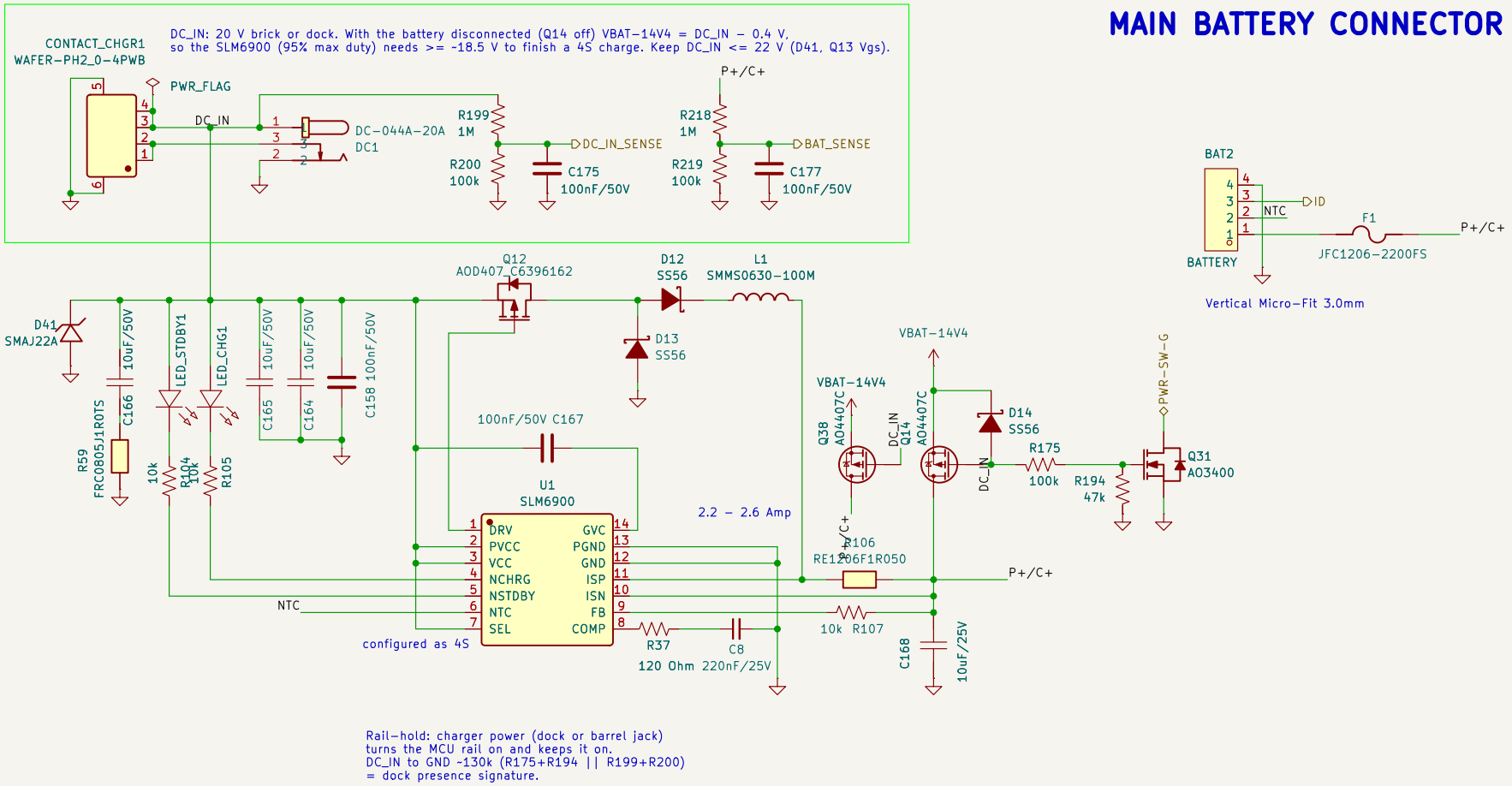


Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

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Size: USLetter | Date: | **Rev:**
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MAIN BATTERY CONNECTOR



Sheet: /BATTERY−CHARGER/		
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